Second-Moment Itô Mass-Flux Tracers in AthenaK

Implementation Decisions, Validation, and Thermodynamic Tracing

Technical note for `feature/ito_tracers`

Implementation baseline: commit `ab2c03880`, June 5, 2026

Abstract

The goal was to add the second-order Itô tracer method to AthenaK as a continuous-path alternative to the existing Monte Carlo mass-flux tracers, while explicitly excluding the third-order method. The implementation derives cell-centered drift and diffusion coefficients from the same final Runge–Kutta-weighted mass fluxes used by the Monte Carlo tracer, communicates those coefficients across MeshBlocks and refinement boundaries, interpolates them to each particle, and advances particles with an Euler–Maruyama step.

The evidence supports a precise conclusion: the implemented Itô-2 particles match the Monte Carlo tracer's per-coordinate displacement mean and variance in controlled transport tests, preserve continuous subcell positions through two-dimensional and AMR motion, and provide useful mass-weighted temperature histories in a two-phase thermal-instability calculation. The evidence does not establish a third-order method, a converged thermal-fragmentation solution, or support outside the explicitly validated non-relativistic, periodic, two- and three-dimensional configurations.

1 What We Were Trying to Accomplish

Monte Carlo mass-flux tracers are tied directly to finite-volume mass exchange: particles jump between cell centers with probabilities derived from the mass fluxes. That makes their ensemble transport physically aligned with the discrete fluid update, but individual paths are discontinuous and cell-centered. For thermodynamic tracing, those jumps can obscure the continuous history one would like to associate with a moving mass element.

The requested feature was therefore narrow:

1. build on the existing Monte Carlo tracer method rather than introduce an unrelated passive-particle model;
2. implement only the second-order Itô method from the method described by Moseley, Teyssier, and Abel (arXiv:2604.23041v2);
3. preserve the Monte Carlo process's first two per-coordinate displacement moments while producing continuous particle paths; and
4. verify that the particles can record meaningful thermodynamic histories in a coupled cooling problem.

2 The Simple Picture

For one cell and one coordinate direction, the Monte Carlo tracer has three possible outcomes over a fluid timestep: jump left, stay, or jump right. The Itô-2 tracer replaces that discrete draw with a continuous random displacement chosen to have the same mean and variance.

If p_R and p_L are the outward Monte Carlo jump probabilities, define

$$C_+ = p_R + p_L, \qquad C_- = p_R - p_L. \tag{1}$$

For cell width h and timestep Δt, the implemented drift and diffusion coefficients are

$$u = \frac{hC_-}{\Delta t}, \qquad \kappa = \frac{h^2\left(C_+ - C_-^2\right)}{2\Delta t}. \tag{2}$$

The particle then advances as

$$X^{n+1} = X^n + u\Delta t + \sqrt{2\kappa\Delta t}\,\xi, \tag{3}$$

where ξ is a bounded uniform random variable with zero mean and unit variance. In the implementation, independent draws are used in each active coordinate direction.

This is the part that matters: the particles still take their transport statistics from the finite-volume mass flux, but they no longer have to sit on cell centers or move by discrete one-cell jumps.

3 Design Choices and Tradeoffs

Approach	Strongest advantage	Main weakness	Decision
Keep only discrete Monte Carlo tracers	Already follows the finite-volume mass exchange and has mature seeding, migration, restart, and output paths.	Individual histories remain cell-centered and discontinuous.	Retained as the baseline.
Use ordinary velocity-following passive particles	Simpler continuous trajectories.	Does not derive transport from the numerical mass flux and therefore does not preserve the Monte Carlo tracer's mixing statistics.	Rejected.
Build Itô-2 on the Monte Carlo infrastructure	Continuous paths with the same per-coordinate mean and variance as the mass-flux jump kernel; reuses established particle infrastructure.	Requires new coefficient fields, MeshBlock/AMR communication, and inherits limits of the existing migration path.	Implemented.
Implement Itô-3	Intended to match additional information beyond second moments.	Explicitly outside the requested scope because of unresolved correctness concerns.	Rejected and fail-closed.

Table 1: The implementation choices were driven by the requirement that continuous particles remain tied to the finite-volume mass transport.

The thermal-instability problem followed the same conservative approach. It uses the existing `cooling_test` problem generator, the standalone ISM cooling/heating module, and the existing initial-perturbation module. Density-only 1% fluctuations were chosen instead of temperature or pressure noise because they preserve a uniform initial temperature while cleanly triggering the density dependence of cooling.

4 What Was Implemented

The new Itô path is integrated into the existing Lagrangian mass-tracer system rather than implemented as a separate particle subsystem.

1. Hydro and MHD accumulate mass flux using the weights that contribute to the final RK1, RK2, or RK3 state. A simple arithmetic average of stage fluxes would not represent the final finite-volume update.
2. After the fluid integrator, each cell converts its outward mass-flux probabilities into u and κ. Invalid probabilities, negative variance beyond tolerance, and non-finite values terminate the run.
3. The coefficient fields are restricted, communicated, and prolonged across MeshBlocks and AMR boundaries.
4. Cloud-in-cell interpolation evaluates u and κ at each particle position.
5. A deterministic hash of tracer tag, cycle, random seed, and coordinate produces the bounded uniform kicks for the Euler–Maruyama update.
6. The existing mass-tracer seeding, migration, restart, VTK output, and thermodynamic-history paths handle the resulting particles.

The implementation also makes its scope explicit. Inputs requesting `pusher=ito3`, an `ito_order` other than 2, or a kick distribution other than the bounded uniform Itô-2 draw fail immediately. The current path requires non-relativistic Cartesian Hydro or MHD, dynamic RK1–RK3 evolution, periodic boundaries in every active direction, and a two- or three-dimensional mesh.

5 Evidence from Controlled Transport Tests

5.1 One-dimensional-style moment test

Because the particle module requires a two- or three-dimensional mesh, the one-dimensional-style test confines motion to x_1 on a 128×4 sheet. It starts 4096 tracers in one x_1 cell and follows them for 64 RK2 steps.

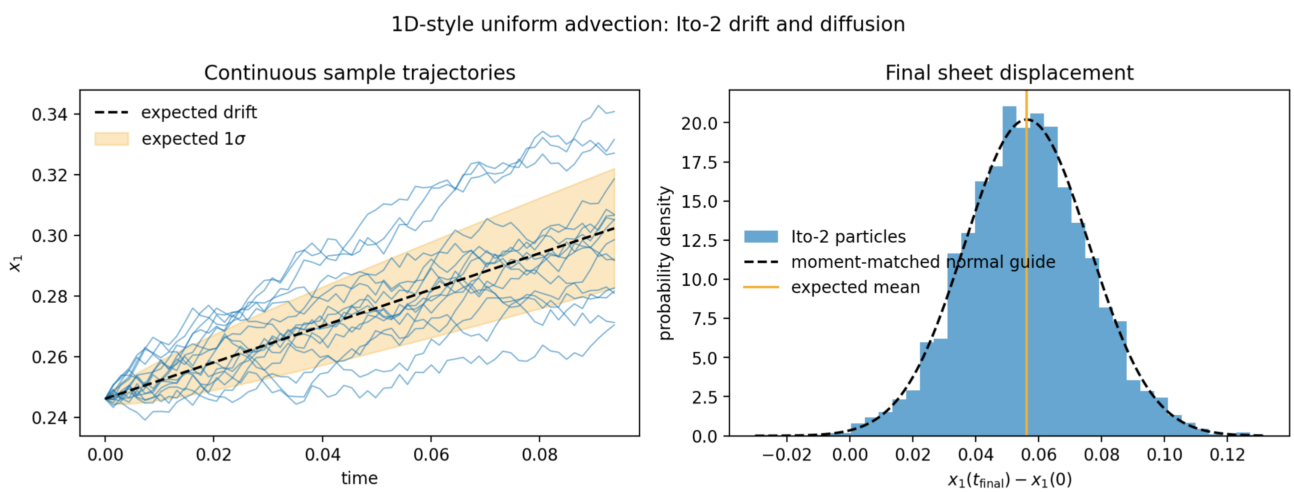

Figure 1: Committed one-dimensional-style behavior test generated by `plot_ito_tracer_figures.py`. The paths are continuous, while the final distribution follows the expected Monte Carlo mean and variance. The measured final mean displacement is 0.05616, compared with the expected 0.05625. This supports per-coordinate moment matching; it does not claim that the continuous distribution reproduces every higher moment of the discrete jump process.

The CPU regression computes the expected mean $\langle \Delta x \rangle = v\Delta t$ and variance $h^2 C(1-C)$ for uniform advection, then checks 4096-particle samples against those values. It also checks restart behavior and verifies that Itô-3 and invalid particle-type combinations are rejected.

5.2 Two-dimensional continuous paths

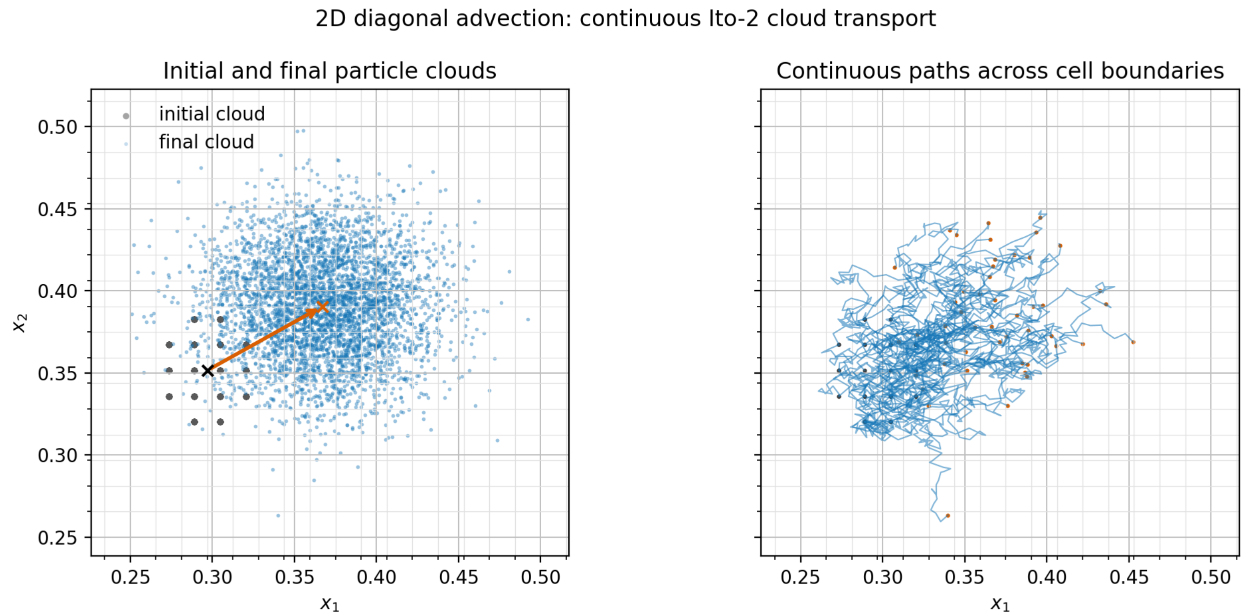

Figure 2: Committed two-dimensional cloud test generated by `plot_ito_tracer_figures.py`. The final cloud both advects and spreads, and the sample paths remain continuous rather than snapping between cell centers. This is direct evidence for the intended geometric behavior, but it is not by itself a test of thermodynamic sampling or AMR communication.

The two-dimensional cloud uses 4096 tracers on a 64×64 mesh with diagonal velocity $(v_1, v_2) = (0.45, 0.25)$. Separate MPI/AMR regression coverage checks that particles migrate across ranks, preserve continuous subcell positions through refinement boundaries, and restart from per-rank restart files.

6 Thermodynamic Tracing of Thermal Instability

6.1 Physical setup

The more informative integration experiment combines the Itô-2 particles with ISM-like cooling and initial perturbations in a periodic $20\,\mathrm{pc} \times 20\,\mathrm{pc}$, 128^2 Hydro calculation. The gas starts in exact heating-cooling balance at

$$\frac{P}{k_B} = 3000\ \mathrm{K\,cm^{-3}}, \qquad n_0 = 1.9071755\ \mathrm{cm^{-3}}, \qquad T_0 = 1573.0068\ \mathrm{K}. \tag{4}$$

This is the unstable middle equilibrium. At the initial pressure, the stable equilibria are near 52 K and 6353 K.

The causal chain is simple. The 1% density-only perturbations leave the initial temperature uniform. Slightly denser regions cool faster because radiative cooling scales as n^2, while constant per-particle heating scales as n. They lose pressure and condense. Slightly underdense regions net heat and expand. The 4096 mass-weighted Itô tracers record which thermal phase their stochastic mass-flux trajectories occupy over time.

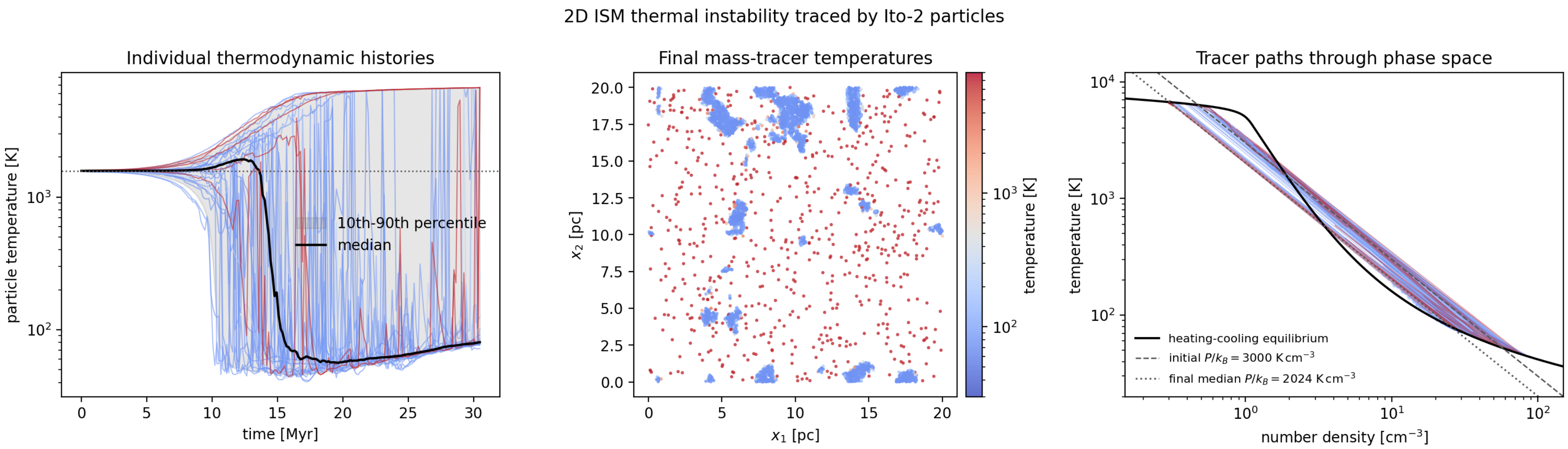

Figure 3: Thermodynamic tracing in the two-dimensional thermal-instability experiment, generated by `plot_ito_thermal_instability.py`. Look first at the left panel: the initially single-temperature population separates into cold and warm histories. The middle panel shows the final mass-tracer temperatures. In the right panel, paths approach the two stable intersections of the heating-cooling equilibrium curve with the evolved median isobar. Sharp jumps in individual histories occur when a particle crosses a thin phase interface because history output samples the fluid cell containing the particle. The figure supports coupled thermodynamic tracing; it does not establish resolution-converged interface structure or fragmentation.

6.2 Observed result

The run completed 100 code time units, or 30.47 Myr, in 9437 cycles. Its thermodynamic-history output contained 823,296 finite records over 201 output cycles for all 4096 tags. At the final output:

- the mass-weighted median pressure was $P/k_B = 2024\ \mathrm{K\,cm^{-3}}$;
- the cooling curve's stable roots at that pressure were 79.5 K and 6720 K, separated by an unstable root at 589 K;
- the measured median tracer temperatures were 79.5 K in the cold phase and 6673 K in the warm phase; and
- the mass-tracer fractions were 81.8% below 300 K, 15.5% above 5000 K, and 2.7% between those thresholds.

The agreement between the measured phase temperatures and the stable heating-cooling intersections is the strongest evidence that the particles are recording the expected thermodynamic evolution. Because the particles were seeded by mass, the reported fractions estimate mass fractions, not volume fractions.

7 Verification and Failure Boundaries

Check	Evidence	Status
Uniform-flow moments	CPU regression compares 4096-particle mean and variance with the analytical Monte Carlo moments; includes RK2 and RK3.	**Passed**
Restart and invalid requests	Serial restart passes; Itô-3, order 3, and wrong particle-type combinations are explicitly rejected.	**Passed**
MHD, MPI, and AMR	Core Itô validation includes MHD thermodynamic fields and a two-rank MPI/AMR migration and restart regression that checks continuous subcell positions.	**Passed for core implementation**
Merged cooling and perturbation integration	Combined serial build and 43 focused Itô, cooling, and initial-perturbation tests.	**Passed**
Thermal-instability experiment	Full 128^2, 4096-particle run completed with all history values finite and phase temperatures consistent with stable equilibria.	**Passed**
Merged-feature MPI integration	No additional MPI run was performed after merging cooling and initial perturbations into the Itô branch.	**Not tested**

Table 2: Verification status. The distinction between core Itô MPI validation and the later merged-feature serial validation is intentional.

The fail-closed checks are part of the implementation, not just the tests. Runs terminate on invalid or non-finite Monte Carlo probabilities, negative diffusion beyond tolerance, invalid interpolated coefficients, non-finite displacements, and displacements that would cross more than one MeshBlock in an active direction.

8 What the Work Supports, and What It Does Not

What is directly supported.

- Itô-2 is integrated with the existing Monte Carlo mass-tracer infrastructure and final finite-volume mass fluxes.
- Controlled tests support the intended per-coordinate first- and second-moment matching and continuous-path behavior.
- The particles can record interpretable mass-weighted thermodynamic histories in a coupled two-phase cooling calculation.

What remains uncertain or explicitly unsupported.

- No Itô-3 or third-moment method is implemented or validated.
- Independent coordinate kicks do not claim to reproduce the full joint discrete jump distribution or all higher moments.
- Current support is limited to non-relativistic Cartesian Hydro/MHD, periodic active boundaries, dynamic RK1–RK3 evolution, and 2D/3D particle meshes.
- The existing particle migration path limits a realized displacement to less than one MeshBlock per coordinate per fluid timestep.
- The thermal-instability experiment omits conduction, so the Field length is unresolved. Interface thickness, small condensations, and phase fractions are resolution-, seed-, domain-, and runtime-dependent.
- The final particle map samples the mass distribution; it is not a complete volume-filling gas-temperature map.

Bottom line. AthenaK now has a deliberately scoped Itô-2 mass-flux tracer that converts the existing Monte Carlo jump statistics into continuous stochastic paths without disconnecting the particles from the finite-volume mass transport. The thermodynamic-tracing use case works, while third-order behavior, unrestricted configurations, full joint-distribution matching, and converged thermal fragmentation remain outside the supported claim.

9 Reproduction and Artifact Map

The thermal-instability experiment and figure can be reproduced from the repository root with:

```
./build/src/athena \
  -i inputs/particles/ito_tracers_thermal_instability_2d.athinput \
  -d run_ito_ti_2d

python scripts/plot_ito_thermal_instability.py \
  run_ito_ti_2d/prtcl_thermo_history/\
ito_tracers_thermal_instability_2d.prtcl_thermo_history.thp
```

Implementation: `particles_lagrangian_ito.cpp`, `driver.cpp`, the Hydro/MHD flux files, and `particles_tasks.cpp`.

Regression evidence: `test_particles_ito_cpu.py` and `test_particles_ito_mpicpu.py`.

Thermal experiment: `inputs/particles/ito_tracers_thermal_instability_2d.athinput`. Figure script: `scripts/plot_ito_thermal_instability.py`.

User documentation: `docs/source/modules/ito_tracers.md`.